

RETIREMENT AND THE DEMOGRAPHIC TRANSITION: A SURVEY EXPERIMENT

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Population is ageing in Italy...

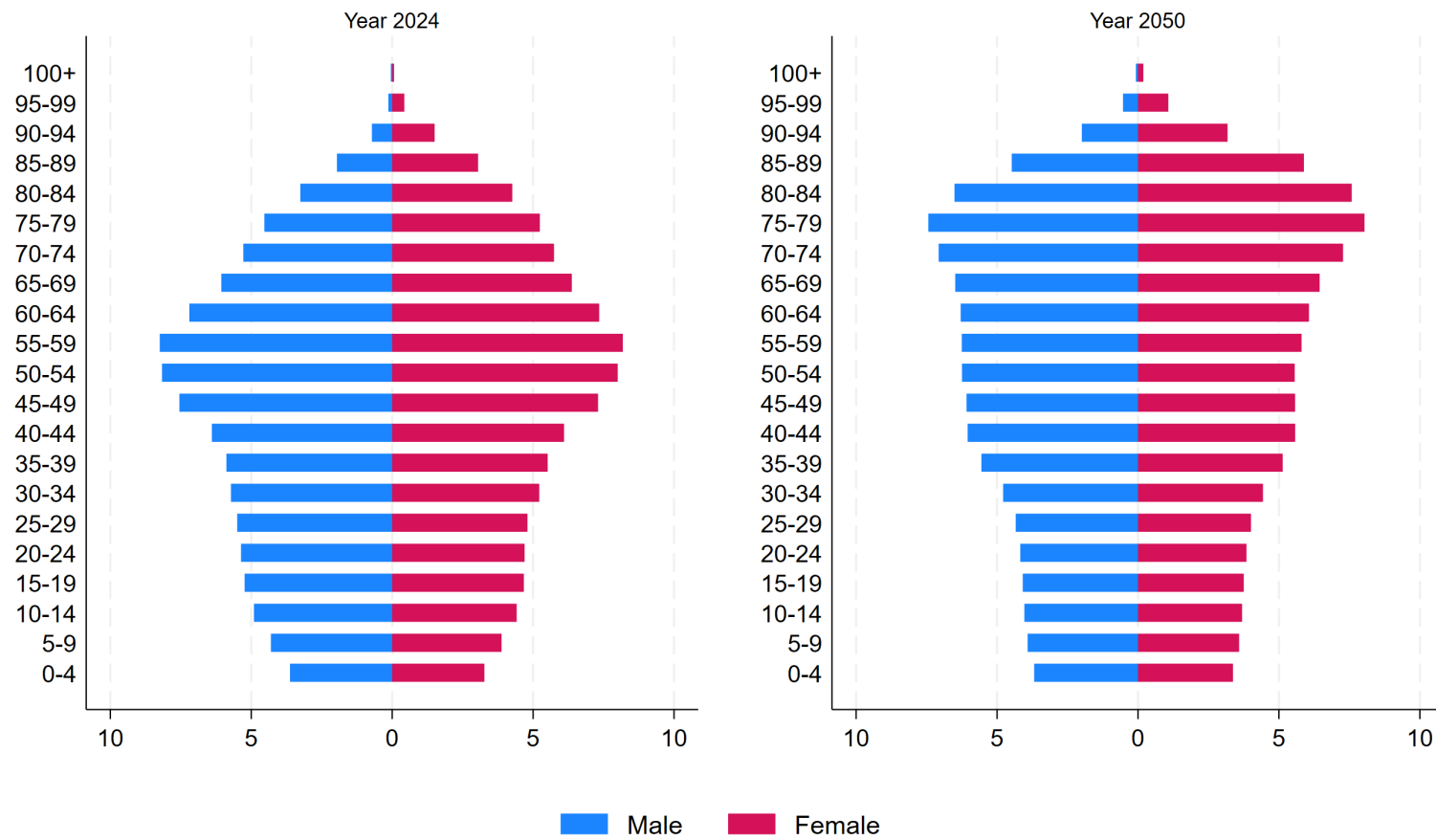


Fig. 1: Share of the male and female population by age in 2024 and 2050. Source: ISTAT

...Major problem for the Pension System

As most advanced countries, Italy runs a pay-as-you-go (PAYGO) scheme:

- current retirees' pensions are paid by current workers

The sustainability of the system hugely depends on demographic trends:

- key is the **Dependency Ratio**: Over 65 / 15-64 Population
- today is around 0.50 and will reach 0.75 in 2070

“Does information on the sustainability issues of the Pension System affect expected social security wealth (SSW)? And do people substitute SSW with alternative private pension plans?”

What I do

- survey experiment** to induce an exogenous variation in information on the Pension System

“According to the latest Annual Report of INPS, the progressive ageing of the population, which increases the number of retirees and reduces the number of active workers, makes it increasingly difficult to finance pensions in all European countries, raising financial stability issues.”

- elicit expected retirement age and replacement rate following **Dominitz and Mansky 2006**
- compute expected SSW as in **Bottazzi et al. 2006**
- look at the effect of the treatment on retirement expectations
- use the treatment as a first stage to estimate the substitutability between expected SSW and private pension plans

Data and Experimental Design

- novel Italian Survey of Consumer Expectations (**Guiso and Jappelli 2024**)
- focus on **workers** younger than 50
- 8 panel** waves of detailed data on demographics, retirement expectations, income, wealth, and composition of financial wealth (on the extensive margin)
- Timing
 - April 2024 (wave 3): the information treatment is delivered
 - right after (still in wave 3): look at the treatment effect on SSW
 - Oct 2024 - 2025 (wave ≥ 4): look at implications on holdings of pension plans

Conclusions and Next Steps

- providing information about Italy's Social Security System **worsens** retirement expectations
- very little** substitutability between the public and private pension funds
- possible theoretical explanations of this result:
 - high transaction costs (**Choukhmane and De Silva 2026**)
 - tail events (**Fagereng, Gottlieb, and Guiso 2017, Daminato and Padula 2024**)
 - low financial literacy (**Jappelli and Padula 2013, Lusardi, Michaud, and Mitchell 2017**)
- on the agenda**: life cycle model of portfolio choice to quantify these frictions
- leverage the empirical estimates as targeted moments in model estimation

Expected Social Security Wealth

- SSW_t is the sum of all annual retirement benefits discounted at t

$$SSW/Y_t = \left[p(N_t|t) \left(\frac{1+g_y}{1+r} \right)^{N_t-t} R_t \right] \left[\sum_{\tau=N_t}^T p(\tau|N_t) \left(\frac{1+g_R}{1+r} \right)^{\tau-N_t} \right]$$

- Where:

- R_t is the exp. replacement rate and N_t the exp. retirement age, t is current age
- Y_t is labour earnings
- $p(\cdot|\cdot)$ are conditional survival probabilities (available from ISTAT)
- T is the terminal age (assumed to be 119)
- $g_y \in \{0, 0.001, 0.005\}$, $g_R = 0$, $r = 0.03$ (annual rates)

- Theory says that if $SSW/Y \uparrow \Rightarrow Wealth/Y \downarrow$ (**Attanasio and Brugiavini 2003**)

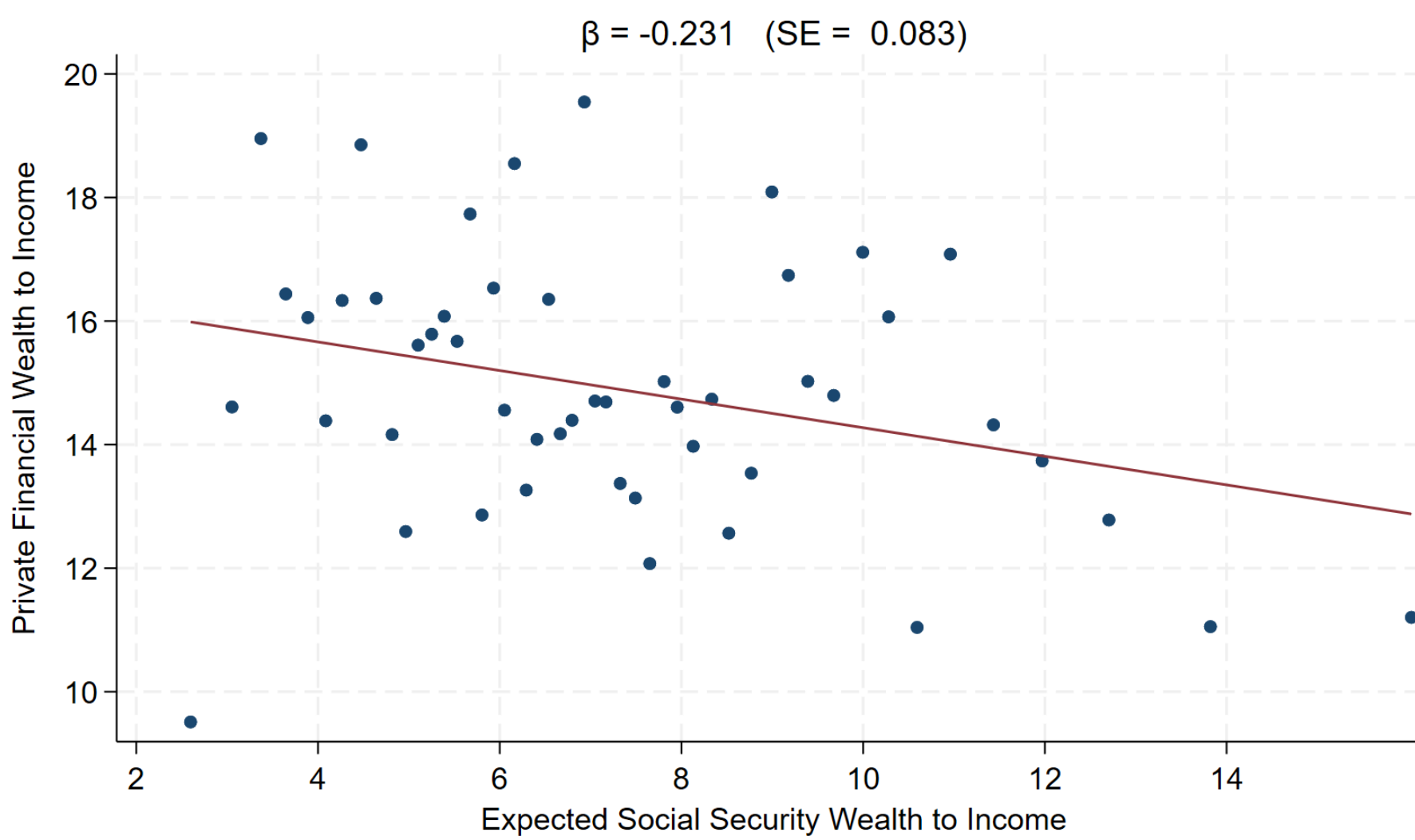


Fig. 2: Correlation between social security wealth (x-axis) private financial wealth (y-axis)

The Effect of the Treatment on SSW

I estimate the following equation:

$$Y_i = \alpha + \beta Treat_i + \varphi X_i + \varepsilon_i \quad (1)$$

- $Y_i \in \{\mathbb{E}_3[Ret\ Age_i], \mathbb{E}_3[Rep\ Rate_i], \mathbb{E}_3[SSW/Y_i], \mathbb{E}_3[SSW_i]\}$
- X_i includes, gender, civil status, family size, employment sector and financial literacy
- β is the coefficient of interest: the ATT on retirement expectations

Expected:	Retirement Age	Replacement Rate (x 100)	SSW/Y	SSW (in €1000)
Treat	0.45** (0.20)	-1.82** (0.84)	-0.44*** (0.15)	-14.61** (7.27)
Controls	✓	✓	✓	✓
Observations	1450	1450	1450	1450

Robust standard errors in parentheses clustered at the individual level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

SSW and Pension Plans ownership

I then estimate the following equation with IV:

$$Private\ Pension_{it} = \alpha + \lambda_t + \gamma \mathbb{E}[SSW_{it}] + \rho X_i + v_{it} \quad (2)$$

- $Private\ Pension_{it}$ is a dummy for holding a private pension plan (x 100)
- $\mathbb{E}[SSW_{it}]$ is instrumented by $Treat_i$ (equation 1)
- λ_t are quarter FEs (starting from wave 4)
- γ is the LATE of $\mathbb{E}[SSW_{it}]$ on $Private\ Pension_{it}$

Sample:	All	Employees	Non holders
Exp. SSW (in €1000)	-0.13 (0.08)	-0.34** (0.15)	0.06 (0.04)
Observations	4960	4313	3390
Baseline Mean (in %)	30.03	31.24	.
Controls	✓	✓	✓
F-stat 1st stage	22.98	12.44	30.19

Robust standard errors in parentheses clustered at the individual level.

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